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Big Data Analysis

Big data $\rightarrow$ high complexity

Propositions $\xrightarrow[\text{linear algebra, calculus}]{\text{sample / Data}}$ Inference $\left\{ \text{Statistics ①} \right.$

Population $\xrightarrow[\text{linear algebra, calculus}]{\text{Big data}}$ Inference $\left\{ \text{Big data analysis ②} \right.$

High performance computing
 $\Downarrow$
 computational linear algebra

③ Law of large numbers

$$x_1, x_2, x_3, \dots, x_m \sim \text{iid}$$

with expected value μ

Then

$$\frac{x_1 + x_2 + \dots + x_m}{m} \rightarrow \mu \rightarrow \text{Number / Vector of number}$$

$\downarrow$
 not necessarily
 1-D, can be a vector

④ Central limit theorem

$$x_1, x_2, \dots, x_m \rightarrow \text{iid} + \text{some condition}$$

$$\& E(x_i) = \mu$$

$$\text{var}(x_i) = \sigma^2$$

sample mean $\frac{x_1 + x_2 + \dots + x_n}{n} - \mu \rightarrow N(0,1)$

$\frac{\sigma / \sqrt{n}}$

⑤

$$\frac{x_1 + x_2 + \dots + x_n - \mu}{\sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}}$$

$\approx t_{n-1} \text{ df}$

$\hookrightarrow$ if n becomes very large t will converge to $N(0,1)$

for $n \gg 0$

$$\frac{\bar{x} - \mu}{\sqrt{s^2/n}} \rightarrow N(0,1)$$

⑥ Dimension $\gg 0$

$\hookrightarrow$ Big data analysis

⑦ generate n independent uniform $[0,1]^n$ & ~~not~~ the pairwise euclidean distances

$$\begin{pmatrix} \vdots \\ \vdots \\ \vdots \end{pmatrix}_{n \times n}$$

$x_1, x_2, \dots, x_n \sim \text{iid uniform}(0,1)$

then $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \sim \text{unif}[0,1]^n$

Make $n \rightarrow \infty$

① Make 5 observations each

② Try to reason
start with $n=2$

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Problem $X_1, X_2, \dots, X_n$ iid $U[0,1]$

$X_{(r)}$ is the r^{th} order statistics i.e.

$$X_{(n)} = \max\{X_1, X_2, \dots, X_n\}$$

$$X_{(1)} = \min\{X_1, \dots, X_n\}$$

$$X_{(1)} \leq X_{(2)} \leq X_{(3)} \leq \dots \leq X_{(n-1)} \leq X_{(n)}$$

Find cdf & pdf fn. of $X_{(r)}$

Challenge problem

Find cdf & pdf of $X_{(n)} - X_{(1)}$

↓
sampling range

(Before next
monday)

Basic Statistics

① Descriptive statistics

↳ not making any formal inference

Data $X_1, X_2, \dots, X_n$

central tendencies : Mean, mode, median, variance etc.
sampling range

Bivariate data

$(X_1, Y_1), \dots, (X_n, Y_n)$

To guess a relation between X & Y

Simple Linear Regression

$Y \approx a + bx$ based on $\{(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)\}$

Find a, b such that

$\ell(a, b) = \sum_{i=1}^n (y_i - a - bx_i)^2$ is minimised

$$\frac{\partial \ell}{\partial a} = 0 \Rightarrow -2 \sum (y_i - a - bx_i) = 0$$

$$\frac{\partial \ell}{\partial b} = 0 \Rightarrow -2 \sum (x_i)(y_i - a - bx_i) = 0$$

$$\sum_{i=1}^n y_i - a - bx_i = 0$$

$$\sum x_i (y_i - a - bx_i) = 0$$

$$\hat{a} = \bar{y} - \hat{b} \bar{x}$$

$$\hat{b} = \frac{\overline{xy} - \bar{x}\bar{y}}{\overline{x^2} - \bar{x}^2}$$

Multiple Linear Regression

Data	$(x_{11}, x_{12}, \dots, x_{1n}, y_1)$	$\left\{ \begin{array}{l} \text{Find} \\ y \approx A\underline{x} + \underline{b} \end{array} \right.$
	$(x_{21}, x_{22}, \dots, x_{2n}, y_2)$	
	$\vdots$	
	$(x_{m1}, x_{m2}, \dots, x_{mn}, y_m)$	

$$y \approx a + b_1 x_1 + b_2 x_2 + \dots + b_n x_n$$

$$\ell(a, b_1, \dots, b_n)$$

$$= \sum (y_i - a - b_1 x_{i1} - \dots - b_n x_{in})^2$$

Minimize ℓ w.r.t

$$a, b_1, \dots, b_n$$

general linear Regression case

$$\underline{Y} \approx A \underline{X} + \underline{b}$$

$$(\underline{Y} | \underline{X})$$

(12) $z_1, z_2, \dots, z_m$ population

$x_1, x_2, \dots, x_m \sim \text{iid from population } Z \rightarrow \text{you pick up}$
note and keep back
as it is

sample mean $\frac{x_1 + x_2 + \dots + x_m}{m} = \bar{x}$

$$E(\bar{X}) = \frac{1}{m} \sum_{i=1}^m E(x_i)$$

$$= \frac{1}{m} \sum_{i=1}^m \sum_{j=1}^m \frac{1}{N} x_i z_j$$

$$= \frac{1}{m} \sum_{i=1}^m \bar{z} = \frac{1}{m} \times m \times \bar{z}$$

$$= \bar{z}$$

$E(\text{sample mean}) = \text{Population mean}$

$$E\left(\frac{1}{m} \sum_{i=1}^m (x_i - \bar{x})^2\right) = \frac{m-1}{m} \cdot \frac{1}{m} \sum_{i=1}^m (z_i - \bar{z})^2$$

$$E\left(\frac{1}{m-1} \sum_{i=1}^m (x_i - \bar{x})^2\right) = \frac{1}{m} \sum_{i=1}^m (z_i - \bar{z})^2$$

sample variance

$x_1, x_2, \dots, x_m$ iid from distribution F_θ

Try to infer about θ based on $x_1, \dots, x_m$

Maximum likelihood method

$$x_1, \dots, x_n \stackrel{\text{iid}}{\sim} \text{Ber}(p)$$

$$P(X_i = x_i) = p^{x_i} (1-p)^{1-x_i}$$

given $x_1, \dots, x_n$

which (p) makes it most likely?

likelihood of $x_1, \dots, x_n$

$$\begin{aligned} \ell(p) &= \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} \\ &= p^{\sum x_i} (1-p)^{n-\sum x_i} \end{aligned}$$

For which p , $\ell(p)$ is maximum?

Taking $\log$ both sides & as $\log$ is increasing function

$$\ln(\ell(p)) = \sum x_i \ln p + (n - \sum x_i) \ln(1-p)$$

$$\frac{d\ell}{dp} = 0$$

$$0 = \frac{\sum x_i}{p} + \frac{(n - \sum x_i)}{(1-p)} (-1)$$

$$\frac{n - \sum x_i}{1-p} = \frac{\sum x_i}{p} \Rightarrow np - \cancel{p \sum x_i} = \sum x_i - \cancel{p \sum x_i}$$

$$p = \frac{\sum x_i}{n} = \bar{x}$$

$p = \bar{x}$

Now, how to prove it is maximum

$$\hookrightarrow \frac{d^2 \ell}{dp^2} < 0 \quad \text{check!!!}$$

19) $X_1, \dots, X_n \stackrel{iid}{\sim} \text{uniform } [0, \theta]$

20) Find maximum likelihood estimate for θ

$$L(\theta) = \prod_{i=1}^n \frac{1}{\theta} \quad X_1, \dots, X_n \leq \theta$$

$$MLE(\theta) = X_{(n)}$$

21) Statistical inference

Population $\xrightarrow[\text{+LA + CALC + comp}]{\text{sample}}$ Inference

22)

Estimation
Hypothesis testing
↓

choice between two statements H_0 & H_1

Reality \ Decision		H_0	H_1	
		H_0	H_1	
H_0	✓	X	→ Type I error	
H_1	X	✓	↓ Type 2 error	

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Uniform problem

1. Variance decreases to zero as dimension increases
2. Mean increases
3. Histogram is bell shaped
4. Peak at mean increases
5. "if n is large then convergence to a stable pattern is quicker"
↳ not a law of large numbers

ex Do the same experiment with Normal $(0, 1)$ truncated at $[-1, 1]$ & exp(1) truncated at $[0, 2]$

Q What is truncated distribution

$g(x) = f(x)$ truncated on $[a, b]$

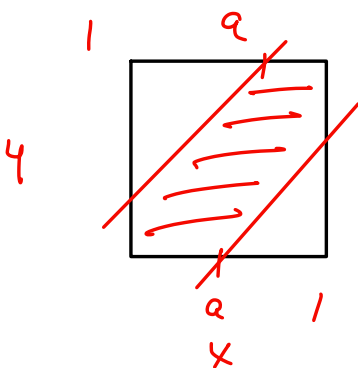
$$\int_a^b f(x) dx = k$$

Then $g(x) = \begin{cases} \frac{f(x)}{k} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$

ex X, Y independent $\sim U[0, 1]$

Find pdf of $|X - Y|$

$F_2(a) : P(|X - Y| \leq a)$



$$F_2(z) = 2z - z^2 \quad 0 \leq z \leq 1$$

$$= 1 \quad z \geq 1$$

$$= 0 \quad z \leq 0$$

$$f_2(z) = \begin{cases} 2-2z & 0 \leq z \leq 1 \\ 0 & \text{o.w} \end{cases}$$

$$\left(\frac{dF_2}{dz} \right)$$

Proof

$(u_{ij})_{m \times m}$ each entry iid $U[0,1]$

$$E(\text{squared Eu dist}) = E\left(\sum_{k=1}^m (u_{ik} - u_{jk})^2\right)$$

$$= \sum_{k=1}^m E(u_{ik} - u_{jk})^2$$

$$= m E(u_{i1} - u_{j1})^2$$

$$= m/6 \quad \forall i, j$$

$$\text{so mean} \approx \sqrt{\frac{\text{dim}}{6}}$$

$d \neq 2$ as iid

K-nearest point estimator

Dependent variable y

Independent " $x_1 \dots x_m$

Data $(x_{11}, x_{12} \dots x_{1m}, y_1)$

$(x_{21}, x_{22} \dots x_{2m}, y_2)$

$\vdots$

$(x_{m1}, x_{m2} \dots x_{mm}, y_m)$

Problem: guess a formula for y based on $x_1 \dots x_m$ based on this data

$$y \approx f(x)$$

For any x the $f(x)$ "should be similar to y_i for nearby $\tilde{x}_i$ "

K-nearest neighbourhood approximation

k +ve integer

Issue $\rightarrow$ as $m \rightarrow \infty$ all points start to appear equidistant

$y = f(x)$ is the average of k -many y_i 's such $\tilde{x}_i$'s are nearest to x

Ex write a code for K-nearest method approximation

If F is any cdf & $Z \sim U[0,1]$ then $F^{-1}(Z) \sim$ distribution of F

Multiple Linear Regression

Data

Want to write $y \approx \beta_0 + \beta_1 x_1 + \dots + \beta_m x_m$

Regression model

Minimise

$$\sum_{i=1}^m (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2} - \dots - \beta_m x_{im})^2$$

with $\beta_0, \beta_1, \dots, \beta_m$ BLUE

Find formula for $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_m$

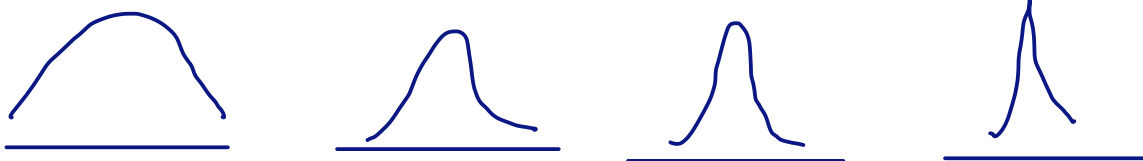
$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_m x_m + \varepsilon$$

$\varepsilon \rightarrow$ is a random variable

$$E(\varepsilon) = 0 \quad \varepsilon \sim N(0, \sigma^2)$$

Uniform simulation problem contd

histogram for squared euclidean distances



Dimension (n) increases

Expected Value $\approx n$

Variance $\approx n$

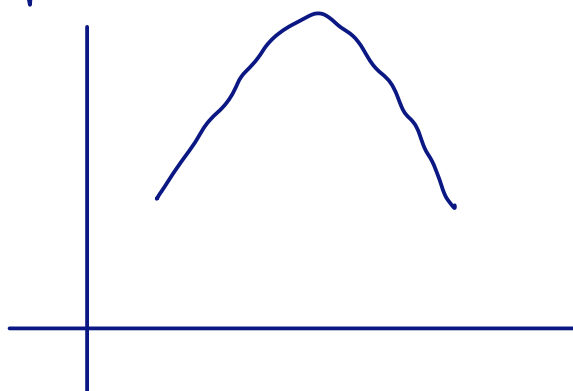
$$\frac{\text{st dev}}{\text{Mean}} \approx \frac{1}{\sqrt{n}} \rightarrow 0 \text{ as } n \rightarrow \infty$$

correction: Variance $\neq 0$

But $\frac{\text{st dev}}{\text{Mean}} \rightarrow 0$

Makes k -nearest neighbourhood estimator hard to use when $n \rightarrow \infty$
as all pair of points appear to be similarly distant !!

Linear Regression



$$Y \approx a + bx$$

$$Y \approx a + bx + cx^2 \quad ??$$

Bias Variance Tradeoff

Models with more parameter fits the given data best but has more variance as abstract var

Ex Find a correct formulation for bias variance tradeoff in case of $Y \approx a + bx$ or $Y \approx a + bx + cx^2$ with 3 data points

Multiple Linear Regression

$(Y, X_1, \dots, X_m) \rightarrow$ Random Variables

We look for

$Y \approx \beta_0 + \beta_1 X_1 + \dots + \beta_K X_K$ that explains a given data

$(Y_1, X_{11}, X_{12}, X_{13}, \dots, X_{1K})$

$(Y_2, X_{21}, X_{22}, \dots, X_{2K})$

$(Y_m, X_{m1}, X_{m2}, \dots, X_{mK})$

best w/rt euclidean distance

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + \epsilon_i$$

$$\underline{Y}_{n \times 1} = \underline{X}_{n \times k} \underline{\beta}_{k \times 1} + \underline{\epsilon}_{n \times 1}$$

Ordinary least square soln to Mult Lin Model

$$\hat{\beta} = \min_B \| \underline{Y} - \underline{X}B \|^2$$

For given data

$$S(\beta) = \sum_{i=1}^n \epsilon_i^2 \\ = \underline{\epsilon}^T \underline{\epsilon}$$

$$(\underline{A}B)^T = B^T \underline{A}^T$$

$$= (\underline{Y} - \underline{X}B)^T (\underline{Y} - \underline{X}B)$$

$$= \underline{Y}^T \underline{Y} + B^T \underline{X}^T \underline{X} B - 2B^T \underline{X}^T \underline{Y}$$

Jacobian & Hessian

$$\frac{\partial S}{\partial B} = 2\underline{X}^T \underline{X} B - 2\underline{X}^T \underline{Y} \quad - (*)$$

$$\frac{\partial^2 S}{\partial B^2} = 2\underline{X}^T \underline{X} \rightarrow \text{+ve definite} \rightarrow \text{soln to } (*) \text{ gives minimum!!}$$

$$\frac{\partial S}{\partial B} = 0$$

$$2\underline{X}^T \underline{X} B = 2\underline{X}^T \underline{Y}$$

$$\hat{B} = (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{Y}$$

→ ordinary least square estimate of β from data

Assumption on $\underline{\varepsilon}$

$$1) \quad E(\underline{\varepsilon}) = 0$$

$$2) \quad E(\underline{\varepsilon}'\underline{\varepsilon}) = \sigma^2 I_n$$

$$3) \quad \text{Rank} \begin{pmatrix} x_{11} & \dots & x_{1k} \\ \vdots & & \vdots \\ x_{n1} & \dots & x_{nk} \end{pmatrix} = k$$

$$4) \quad \underline{\varepsilon} \sim N(0, \sigma^2 I_n)$$

$b = (X^T X)^{-1} X^T Y \rightarrow$ The random variable b is called the least square estimator of β

Estimation error

$$\begin{aligned} & b - \beta \\ &= (X^T X)^{-1} X^T Y - \beta \\ &= (X^T X)^{-1} X^T (X\beta + \varepsilon) - \beta \\ &= (X^T X)^{-1} X^T \varepsilon \end{aligned}$$

Bias

$$\begin{aligned} & E(b - \beta) \\ &= E((X^T X)^{-1} X^T \varepsilon) \\ &= (X^T X)^{-1} X^T E(\varepsilon) \\ &= 0 \end{aligned}$$

least square estimator is unbiased

Gauss Markov Theorem Proof

$$y = X\beta + \varepsilon$$

$\hat{\beta} = (X^T X)^{-1} X^T y$ is the best linear unbiased estimator of β

Let $\tilde{\beta} = Cy$ be another linear unbiased estimator of β

$$C = (X^T X)^{-1} X^T + D$$

$$E(\tilde{\beta}|X) = E(Cy|X)$$

$$= E((X^T X)^{-1} X^T + D)(X\beta + \varepsilon)|X)$$

$$= ((X^T X)^{-1} X^T + D) X\beta$$

$$+ ((X^T X)^{-1} X^T + D) E(\varepsilon|X)$$

$$= ((X^T X)^{-1} X^T + D) X\beta$$

$$= \cancel{(X^T X)^{-1} X^T} X\beta + D X\beta$$

$$(I_k + DX)\beta$$

$\tilde{\beta}$ is unbiased iff $DX = 0$

Note $\sigma^2 (X^T X)^{-1} = \text{Var}(\hat{\beta}|X)$

$$\text{Var}(\hat{\beta}|X)$$

$$= \text{Var}(Cy|X)$$

$$= C \text{Var}(y|X) C^T$$

$$\sigma^2 C C^T$$

$$\sigma^2 ((X^T X)^{-1} X^T + D)(X(X^T X)^{-1} + D^T)$$

$$= \sigma^2 (X^T X)^{-1} (X^T X) (X^T X)^{-1} + (X^T X)^{-1} X^T D^T + D X (X^T X)^{-1} + D D^T$$

$$= \sigma^2 (X^T X)^{-1} + \sigma^2 (X^T X)^{-1} (D X)^T + \sigma^2 D X (X^T X)^{-1} + \sigma^2 D D^T$$

$$= \sigma^2 (X^T X)^{-1} + \sigma^2 D D^T$$

$$= \text{Var}(\hat{\beta}) + \sigma^2 D D^T$$

$D D^T$ is +ve definite

$$\text{so } \text{Var}(\tilde{\beta}|X) \geq \text{Var}(\hat{\beta}|X)$$

"linear estimator in Y "

$$x^3 y + x^5 \text{ lin in } Y!!$$

For any $z \in \mathbb{R}^{n \times 1}$ & any matrix $A_{m \times n}$ if $\text{Var}(z_{n \times 1}) = B_{n \times n}$

$$\text{Var}(AZ) = A \text{Var}(Z) A^T$$

$$= A B A^T$$

$$\text{if } B = \sigma^2 I$$

$$\text{Var}(AZ) = \sigma^2 A A^T$$

$$\text{Var}(\hat{\beta}|X) = \text{Var}((X^T X)^{-1} X^T Y|X)$$

$$= (X^T X)^{-1} X^T \sigma^2 I X (X^T X)^{-1}$$

$$= \sigma^2 I (X^T X)^{-1}$$

Q Take the following situation $(X, Y) \sim \text{BIV NORM } (0, 0, 1, 1, \rho)$

Take $(x_1, y_1), (x_2, y_2), (x_3, y_3)$

Take least square curves for

$$y = \beta_0 + \beta_1 x$$

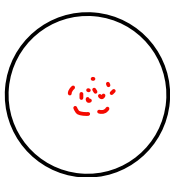
$$y = \gamma_0 + \gamma_1 x + \gamma_2 x^2$$

Then suitable various size 2 samples and compare their behaviour

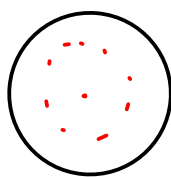
- a parameter is some quantity about a distribution that we would like to know (from some sample)
- We shall estimate θ using an estimate $\hat{\theta}$ (a function of the sample)

$$\text{Bias}(\hat{\theta}, \theta) = E(\hat{\theta}) - \theta$$

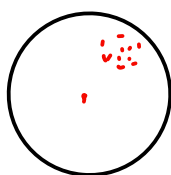
$$\text{Var}(\hat{\theta}) = E(\hat{\theta} - E(\hat{\theta}))^2$$



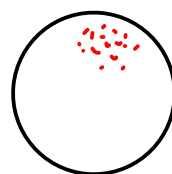
low B
low V



low B
high V



high B
low V



high B
high V

Some Decision Theory

Our data comes from distribution

Δ on $\mathcal{X} \times \mathcal{Y}$ where $\mathcal{Y} \subseteq \mathbb{R}$

We usually do not know Δ but just have the data

$(x_1, y_1), \dots, (x_n, y_n)$

We want to find a function

$f: \mathcal{X} \rightarrow \mathcal{Y}$ that

"best explains" the data that is we want to learn a function

approach

choose f such that

$$E_{x, y \sim \Delta} [y - f(x)]^2 \rightarrow \text{is minimum} \quad \text{--- } \textcircled{\#}$$

least squared error

$\rightarrow$ is minimum

What minimises it?

Then $f^*(x) = E_{y|x}[y|x]$ is the solution to the minimisation problem $\textcircled{\#}$

Proof :

$$\begin{aligned} & E_{y|x} [(y - f(x))^2 | x] \\ &= E_{y|x} [(y - E_{y|x}(y|x) + E_{y|x}(y|x) - f(x))^2 | x] \\ &= E_{y|x} ((y - E_{y|x}(y|x))^2 + E_{y|x}(E_{y|x}(y|x) - f(x))^2 \\ &\quad + 2E_{y|x}(y - E_{y|x}(y|x))(E_{y|x}(y|x) - f(x)) | x) \\ &= E_{y|x} ((y - E_{y|x}(y|x))^2 | x) + E_{y|x} ((E_{y|x}(y|x) - f(x))^2 | x) \\ &\quad + 2(E_{y|x}(y|x) - f(x)) E_{y|x} ((y - E_{y|x}(y|x)) | x) \end{aligned}$$

$$\begin{aligned} E((y - E_{y|x}(y|x)) | x) &= E_{y|x}(y|x) - E_{y|x}(y|x) \\ &= 0 \end{aligned}$$

$$\rightarrow E_{y|x} ((y - E_{y|x}(y|x))^2 | x) + E_{y|x} ((E_{y|x}(y|x) - f(x))^2 | x)$$

$$E_{y|x} ((y - f(x))^2 | x) \geq E_{y|x} ((y - E_{y|x}(y|x))^2 | x)$$

$$\forall f \quad \text{---} \textcircled{*}$$

Taking E_x on both side of $\textcircled{*}$ we get the result we get,

$$E[y - f(x)^2] \geq E[y - E_{y|x}(y|x)^2]$$

but basically useless, as we don't know probability disten's

How to pick f ?

Take $E(y|x)$ if you know but in most cases you don't

Ex Find f that minimizes $E[|y - f(x)|]$

→ Given that Δ is not known, f has to be created using the given data S

$S: (x_1, y_1) \dots (x_e, y_e)$

$$(x_i, y_i) \stackrel{\text{iid}}{\sim} \Delta$$

We want to use S to "learn" a function

$$f_\Delta: x \rightarrow y$$

Let us assume some learning algorithm (an alg that takes S as an input & f_Δ as an output)

Then its expected prediction error

$$E_{x, y, \Delta} [(y - f_\Delta(x))^2]$$

Mean prediction of the algorithm

$$\bar{f}(x) = E_\Delta [f_\Delta(x)]$$

$$E_{\Delta, S|x} [(y - f_\Delta(x))^2 | x]$$

$$E_{y, S|x} [y - E_{y|x}(y|x) + E_{y|x}(y|x) - f_\Delta(x)^2 | x]$$

$\Rightarrow$

$$E_{y|\Delta} (y - E_y(y|x))^2 | x) + E_{S|x} ((E_{y|x}(y|x) - f_\Delta(x))^2 | x)$$

$$+ 2E_{y, S|x} [(y - E_{y|x}(y|x)) (E_{y|x}(y|x) - f_\Delta(x)) | x]$$

$$E_{y|x} \left(\underbrace{y - E_{y|x}(y|x)}_0 \cdot E_{s|x}(E_{y|x}(y|x) - f_s(x) | x) \right)$$

$$= E_{y|x}((y - E_{y|x}(y|x))^2 | x) + E_{s|x}(E_{y|x}(y|x) - f_s(x))^2 | x)$$

look at second term

$$E_{s|x}((E_{y|x}(y|x) - E_s(f_s(x)) + E_s(f_s(x)) - f_s(x))^2 | x)$$

$$= E_{s|x}((E_{y|x}(y|x) - E_s(f_s(x)))^2 | x) \quad \text{--- (2)}$$

$$+ E_{s|x}((E_s(f_s(x)) - f_s(x))^2 | x) \quad \text{--- (3)}$$

$$+ 2E_{s|x} \left[\underbrace{(E_{y|x}(y|x) - E_s(f_s(x)))(E_s(f_s(x)) - f_s(x))}_{=0} | x \right]$$

$$\begin{aligned} E[(y - f_s(x))^2 | x] &= E_y(y - E(y|x))^2 | x + E[E(y|x) - E_s(f_s(x))]^2 | x \\ &\quad + E[(E_s(f_s(x)) - f_s(x))^2] \\ &= E_y[y - E(y|x)]^2 | x \rightarrow \text{Inherent Variability} \end{aligned}$$

Theorem (Bias-Variance)

$$\begin{aligned} E_{y,s|x}((y - f_s(x))^2 | x) &= E_{y|x}(y - E_{y|x}(y|x))^2 \quad \text{(i)} \quad \text{--- Variance} \\ &\quad + E_s(f_s(x) - E_s(f_s(x)))^2 | x \\ &\quad + \underbrace{(E_{y|x}(y|x) - E_s(f_s(x)))^2}_{\text{(ii)}} \quad \text{--- Bias} \end{aligned}$$

✓ fixed X

$$E(y - f_S(x)^2 | x) = \text{Var}_{y|x}(y) + \text{Var}_{S|x}(f_S(x)) + \text{bias}(f_S(x)^2 | x)$$

Exc give an example with 2 estimators f_S & f'_S s/t

$$(ii) f_S > (ii) f'_S$$

$$\& (iii) f_S < (iii) f'_S$$

$$E_{x,y,S}[y - f_S(x)^2] = E_x[\text{Var}_{y|x}(y) + \text{Var}_{S|x}(f_S(x)) + \text{bias}(f_S(x)^2 | x)]$$

Exc Try to come up with a bias dispersion formula for absolute deviation

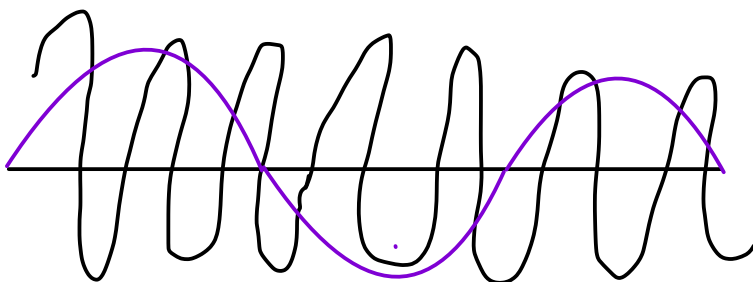
Exc Recall linear regression setup

$$X \sim U[0, 5]$$

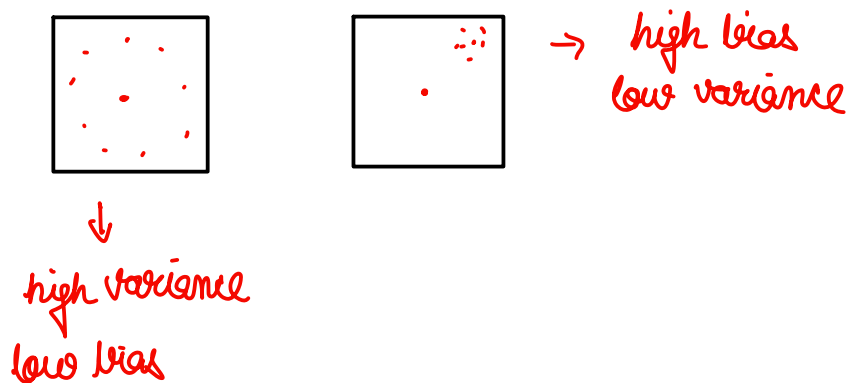
$$\text{Let } Y = \sin(2\pi X) + \varepsilon \quad X, \varepsilon \text{ independent}$$

$$\varepsilon \sim N(0, \sigma^2)$$

Try to do polynomial regression with 3, 4, 5, 6, 7, 8, 9 data points. Find which degree polynomial gives best fit in each case and ~~various~~ very small σ



5/2/24



Multivariate
Regression &
Bias Variance
Tradeoff

Results / Problems

Notation: $x_1, x_2, \dots, x_m$ is an iid sample from an m -dimensional distribution with mean vector $\underline{\mu}$ and variance-covariance matrix Σ

Let S_N be the sample variance-covariance matrix i.e.

$$S_m = \begin{pmatrix} & & \\ & S_{ij} & \\ & & \end{pmatrix}_{m \times m}$$

$$S_{ij} = \frac{1}{n} \sum_{k=1}^n (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)$$

where

$$\begin{pmatrix} x_{11} & \dots & x_{1m} \\ \vdots & & \vdots \\ x_{ni} & \dots & x_{nm} \end{pmatrix} \text{ is a given sample}$$

$(x_{i1} \dots x_{im})$ is the i^{th} data point x_i^T

$$\bar{x}_i = \frac{1}{n} \sum_{k=1}^n x_{ki}$$

Prove

$$E(\bar{x}) = \mu$$

$$\text{var}(\bar{x}) = \frac{1}{n} \sum$$

$$E(S_m) = \sum - \frac{1}{n} \sum$$

} for multivariate

we know $E(\bar{x}) = \mu$

$$\text{var}(\bar{x}) = \frac{\sigma^2}{n}$$

$$E\left(\frac{1}{n} \sum (x_i - \bar{x})^2\right) = \left(1 - \frac{1}{n}\right) \sigma^2$$

} univariate

Proof

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$$E(\bar{x}) = E\left(\frac{x_1 + x_2 + \dots + x_n}{n}\right)$$

$$= \frac{1}{n} \sum_{k=1}^n E(x_k)$$

$$= \frac{1}{n} \sum_{k=1}^n \mu = \mu$$

Q.E.D.

$$(\bar{x} - \underline{\mu})(\bar{x} - \underline{\mu})^T = \left(\frac{1}{n} \sum_{k=1}^n (x_k - \underline{\mu})\right) \left(\frac{1}{n} \sum_{k=1}^n (x_k - \underline{\mu})\right)^T$$

$$\text{cov}(\bar{x}) = E(\bar{x} - \underline{\mu})(\bar{x} - \underline{\mu})^T$$

$$= \frac{1}{n^2} \left(\sum_{k=1}^n \sum_{l=1}^n E(\underline{x}_k - \underline{\mu})(\underline{x}_l - \underline{\mu})^T \right)$$

for $k \neq l$

$$E(\underline{x}_k - \underline{\mu})(\underline{x}_l - \underline{\mu}) = 0 \quad [\text{by independence}]$$

hence

$$\text{cov}(\bar{X}) = \frac{1}{n^2} \left(\sum_{k=1}^n E(\underline{x}_k - \underline{\mu})(\underline{x}_k - \underline{\mu})^T \right)$$

$$\text{As } \Sigma = E(\underline{x}_k - \underline{\mu})(\underline{x}_k - \underline{\mu})^T \rightarrow \text{Main point that}$$

we have

variance-covariance matrix is this

$$\text{cov}(\bar{X}) = \frac{1}{n^2} \left(\sum_{k=1}^n E(\underline{x}_k - \underline{\mu})(\underline{x}_k - \underline{\mu})^T \right)$$

$$= \frac{1}{n^2} (n \Sigma)$$

$$= \frac{1}{n} \Sigma$$

generalised variance

$$S = \begin{bmatrix} s_{11} & - & - & - & s_{1m} \\ \vdots & & & & \\ s_{m1} & - & - & - & s_{mm} \end{bmatrix} \rightarrow \text{sample variance covariance matrix}$$

$|S|$ is called the generalised variance

Result: generalised variance is zero if and only if at least one row of Δ is lin comb of others

$$\Delta = \begin{bmatrix} x_{11} - \bar{x}_1 & x_{12} - \bar{x}_2 & - & - & - & x_{1m} - \bar{x}_m \\ x_{21} - \bar{x}_1 & x_{22} - \bar{x}_2 & & & & \vdots \\ & \vdots & & & & \vdots \\ & & & & & \vdots \end{bmatrix}$$

$$X = \begin{pmatrix} x_{11} & & x_{1m} \\ \vdots & & \vdots \\ x_{m1} & & x_{mm} \end{pmatrix} \begin{pmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_m^T \end{pmatrix}$$

$x_1, x_2, \dots, x_m$ iid are from some common distribution i.e. a sample from some common population

Population mean μ

" Var cov matrix Σ

$$\begin{array}{c} S_m \\ \downarrow \\ \text{sample variance} \end{array} = \begin{pmatrix} S_{ij} \end{pmatrix}_{m \times m} \quad \bar{x}_i = \frac{1}{n} \sum_{k=1}^n x_{ki} \quad \forall i = 1, \dots, m$$

$$S_m = \frac{1}{n} \sum_{k=1}^n (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)$$

$$\bar{x} = (\bar{x}_1, \bar{x}_2, \dots, \bar{x}_m)_{1 \times m}$$

Result

$$E(X) = \mu$$

$$\text{cov}(\bar{x}) = \frac{1}{n} \Sigma$$

(Var cov matrix of $\bar{x}$)

$$E(S_m) = \Sigma - \frac{1}{n} \Sigma$$

$$(y_1, \dots, y_m) \sim L$$

$$\text{Then } \Sigma_{ij} = E(y_i - E(y_i))(y_j - E(y_j))$$

$$S_{ij} = \frac{1}{n} \sum_{k=1}^n (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)$$

$$E(S_m) = \frac{n-1}{n} \Sigma$$

$$E\left(\frac{n}{n-1} S_m\right) = \Sigma$$

$S \rightarrow$ unbiased sample variance

$$S_{ij} = \frac{1}{n} \sum_{k=1}^n (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)$$

$$E(S_{ij}) = \frac{1}{n} E\left(\sum_{k=1}^n (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)\right)$$

$$= \frac{1}{n} E\left(\sum_{k=1}^n (x_{ki} - \mu_i + \mu_i - \bar{x}_i)(x_{kj} + u_j - u_j - \bar{x}_j)\right)$$

$$= \frac{1}{n} \sum_{k=1}^n E\left[\begin{aligned} &(x_{ki} - \mu_i)(x_{kj} - \mu_j) + (x_{ki} - \mu_i)(u_j - \bar{x}_j) \\ &+ (\mu_i - \bar{x}_i)(x_{kj} - \mu_j) + (\mu_i - \bar{x}_i)(u_j - \bar{x}_j) \end{aligned} \right]$$

$$= \frac{1}{n} \left[\sum_{ij} y_{ij} + \frac{\sum_{ij} y_{ij}}{n} \right] + \sum_{k=1}^n \underbrace{(x_{ki} - \mu_i)(u_j - \bar{x}_j)}_{n \frac{\sum_{ij} y_{ij}}{n}} + \underbrace{(x_{kj} - \mu_j)(\mu_i - \bar{x}_i)}_{-n \frac{\sum_{ij} y_{ij}}{n}}$$

$$= \frac{1}{n} [n \sum_{ij} y_{ij} + \sum_{ij} y_{ij} - 2 \sum_{ij} y_{ij}]$$

$$= \frac{1}{n} [n-1] \sum_{ij} y_{ij}$$

$$= \left(1 - \frac{1}{n}\right) \sum_{ij} y_{ij}$$

$$D = X - I_{m \times 1} \bar{X}_{1 \times m}^T$$

Result 2: $|S|=0$ if and only if at least one col of D is lin comb of others (Exercise)

Result 3: If sample size is $\leq$ dimension then $|S|=0$

Q Let $x_1, \dots, x_n$ be uniform inside unit n -dim sphere. What is

$$\lim_{n \rightarrow \infty} P(|x_n| > \epsilon) \text{ for some } 0 < \epsilon < 1 \quad \rightarrow \quad P(x > \epsilon) = 2 \frac{(1-\epsilon)}{2} = 1-\epsilon$$

Multivariate Normal

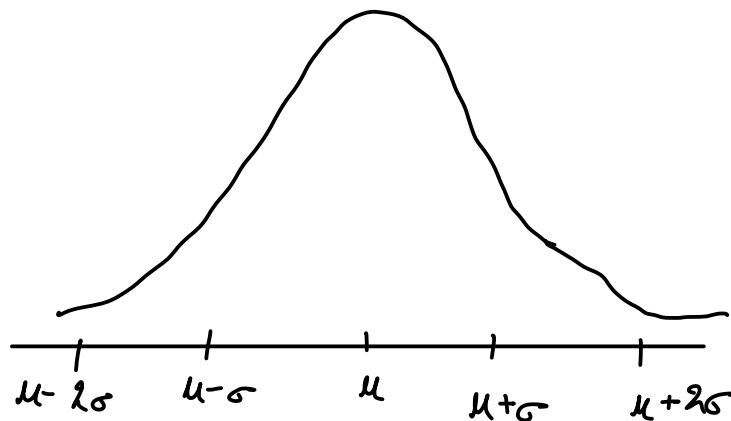
single variable $N(\mu, \sigma^2)$ density $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

Expected value μ

Variance σ^2

$$-\infty < x < \infty$$

$$X \sim N(\mu, \sigma^2)$$



$$P(\mu - \sigma < X < \mu + \sigma) = 0.68$$

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.95$$

Let $\underline{x}_{m \times 1}$ be an m dimensional vector

$$\underline{x} \sim N_m(\underline{\mu}_{m \times 1}, \Sigma_{m \times m})$$

$$(x_1, \dots, x_m)^T$$

if x has density

$$f(x) = \frac{1}{(2\pi)^{m/2} |\Sigma|^{1/2}} e^{-\frac{(\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu})}{2}}$$

Mult Var Normal

$$\text{Hence } E(\underline{x}) = \underline{\mu}$$

$$\text{cov}(\underline{x}) = \Sigma$$

Result 1: Σ is +ve definite (hence Σ^{-1} exists)

Result 2: $X \sim N_m(\underline{\mu}, \Sigma)$ if & only if $\forall$ vector $\underline{a}^T$

$$\underline{a}^T \underline{X} = a_1 x_1 - \dots - a_m x_m \\ \sim N(a^T \underline{\mu}, \underline{a}^T \Sigma \underline{a})$$

Result 3: $X \sim N_m(\underline{\mu}, \Sigma)$

$$A_{m \times m} X = \begin{bmatrix} a_{11} x_1 + \dots + a_{1m} x_m \\ \vdots \\ a_{m1} x_1 + \dots + a_{mm} x_m \end{bmatrix} \\ \sim N_m(A \underline{\mu}, A \Sigma A')$$

Also $X + \underline{d}_{m \times 1}$

distribution as $N_m(\underline{\mu} + \underline{d}, \Sigma)$

Result 4: $X \sim N_m(\underline{\mu}, \Sigma)$

Then for $X = \begin{bmatrix} \frac{X_1, q \times 1}{X_2, m-q \times 1} \end{bmatrix}$

$$\underline{\mu} = \begin{bmatrix} \frac{\mu_1, q \times 1}{\mu_2, m-q \times 1} \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \Sigma_{11, q \times q} & \Sigma_{12, q \times (m-q)} \\ \hline \Sigma_{21, (m-q) \times q} & \Sigma_{22, (m-q) \times (m-q)} \end{bmatrix}$$

we have $X_1 \sim N_q(\underline{\mu}_1, \Sigma_{11})$

Result 5

a) If $X_1 \sim_{q_1 \times 1}$ & $X_2 \sim_{q_2 \times 1}$ are independent then $\text{cov}(X_1, X_2) = \underline{0}_{q_1 \times q_2}$

⑤ For mult normal: X_1 & X_2 indep.
iff $\Sigma_{12} = \underline{0}$

Result 6

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim N_m(\mu, \Sigma)$$

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

$$\& |\Sigma_{22}| > 0 \text{ (p-def)}$$

Then the conditional distribution of X_1 given $X_2 = x_2$ is

$$N_p(\underline{\mu}_1 + \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2), \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21})$$

Theorem : Central Limit Theorem

$X_1, X_2, \dots$ be sequence of independent r.v. such that

$$E(X_i) = \mu$$

$$\text{Var}(X_i) = \sigma^2 < \infty$$

$$\text{Define } U_n = \frac{X_1 + \dots + X_n - n\mu}{\sigma/\sqrt{n}}$$

Then $U_1, U_2, \dots$ converges in distribution to $N(0, 1)$

Recall : $X_1, \dots, X_n$ converges in distr. to X if

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x) \quad \forall x$$

Then (Weak law of large Numbers)

$$X_1, \dots, X_n \text{ iid for } E(X_i) = \mu$$

$$\& \text{Var}(X_i) = \sigma^2 < \infty$$

$$\text{Let } \bar{X}_n = \frac{X_1 + \dots + X_n}{n}$$

Then $\forall \epsilon > 0$

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| < \epsilon) = 1$$

i.e. $\bar{X}_n \rightarrow \mu$ in probability

proof

if g is non neg fn

$$E(g(x)) = \int_{-\infty}^{\infty} g(x) f_x(x) dx$$

$$\geq \int_{x: g(x) \geq k} g(x) f_x(x) dx$$

$$\geq \int_{x: g(x) \geq k} k f_x(x) dx$$

$$= k P(g(x) \geq k)$$

$$\text{so } P(g(x) \geq k) \leq \frac{E(g(x))}{k}$$

$$\text{Hence } P(|\bar{X}_n - \mu| < \epsilon)$$

$$= 1 - P(|\bar{X}_n - \mu| \geq \epsilon)$$

$$\geq 1 - \frac{E(\bar{X}_n - \mu)^2}{\epsilon^2}$$

[can do for discrete similarly]

$$= 1 - \frac{\text{Var}(x_n)}{\varepsilon^2}$$

$$= 1 - \frac{\sigma^2}{n\varepsilon^2} \rightarrow 1 \text{ as } n \rightarrow \infty$$

Questions for practice

1. Give example of a pairs of random variables $(X, Y), (Z, W)$ st $E(Y) = E(W)$ $Var(X) = Var(Z)$ $Var(Y) = Var(W)$ $S(X, Y) = S(Z, W)$ but joint distributions of (X, Y) & (Z, W) are different
2. State & prove Gauss Markov theorem. Give an example of an unbiased estimator of μ (in context of GLM) that is linear in X yet not BLUE. Explain
3. What is weak law of large numbers. Prove it
4. Give an example of two uniform random variables X & Y st (X, Y) is not uniform
5. What is bias-variance tradeoff? Explain with explicit examples
6. If X & Y are normals then must the joint distribution (X, Y) be bivariate normal always?
7. Let $\underline{X} \sim N_{m_1}(\underline{\mu}_1, \Sigma_1)$ & $\underline{Y} \sim N_{m_2}(\underline{\mu}_2, \Sigma_2)$ be two mutually independent random normal vectors. What is the distribution of $(\underline{X}, \underline{Y})$?
8. a) If X & Y are continuous random variables with pdf's f_X, f_Y then is it possible to find a fn. st $h(X, Y)$ has a discrete distribution
 b) If $X_1, \dots, X_m$ are iid $U[0, 1]$ then find distribution of $X_{(n)}$ where $X_{(n)}$ is the n^{th} entry where $X_1, \dots, X_m$ is put in the increasing order

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)} \leq \dots \leq X_{(m)}$$

 c) Show that as n goes to infinity the volume of unit sphere tends to be concentrated on its surface
- 9) Let $\underline{X} \sim N_3(\underline{\mu}, \Sigma)$ where $\underline{\mu}^T = (-3 \ 1 \ 4)$

$$\Sigma = \begin{pmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

 which of the following are independent? In each case find the joint density?
 a) X_1 & X_2 b) X_2, X_3 c) X_1, X_2 d) $\frac{X_1 + X_2}{2}$ & X_3
 e) X_2 & $X_2 - \frac{5}{2}X_1 - X_3$
- 10) Let $X_1, X_2, X_3, X_4 \sim iid \ N(0, 1)$ Find joint densities of
 a) $X_1^2 + X_2^2 + X_3^2 + X_4^2$
 b) $(X_1 - \frac{X_1 + X_2}{2})^2 + (X_2 - \frac{X_1 + X_2}{2})^2$

Q Let x_n be uniform inside unit n -dim sphere. What is

$$\lim_{n \rightarrow \infty} P(|x_n| > \epsilon) \text{ for some } 0 < \epsilon < 1 \rightarrow P(x > \epsilon) = \frac{2(1-\epsilon)}{2} = 1-\epsilon$$

$$P(x_2 > \epsilon) = \frac{\pi(1^2 - \epsilon^2)}{\pi 1^2} = 1 - \epsilon^2$$

$$P(x_3 > \epsilon) = \frac{1^3 - \epsilon^3}{1^3} = 1 - \epsilon^3$$

$$P(x_n > \epsilon) = 1 - \epsilon^n$$

$$\lim_{n \rightarrow \infty} P(x_n > \epsilon) = \lim_{n \rightarrow \infty} (1 - \epsilon^n) = 1$$

for a larger dimension probability that (x_n) will be greater than ϵ is 1